

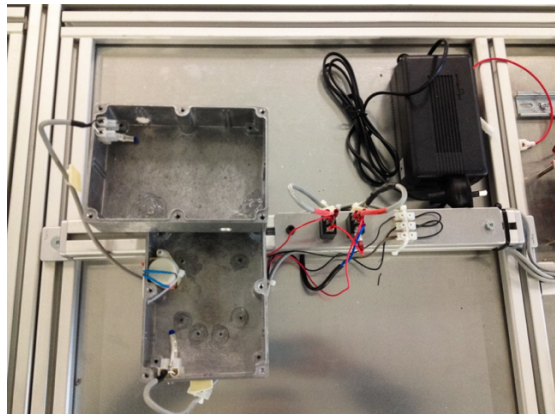


UNIVERSITY  
OF WOLLONGONG  
AUSTRALIA

# LAB REPORT

Subject: MECH470/970

## Water Heating and Pumping Station



Week 6

Group 1  
Team A

| Student Name      | Student Number |
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## 1. Objective

The objective of this lab is to design the ladder logic diagram for a Water Heating and Pumping Station. The apparatus includes two tanks: the heating tank (lower tank) and the process tank (upper tank). The water is to be heated to a maximum of 40°C and be pumped to the process tank, where it is stored at 30°C with an error of 2°C.

Two analogue inputs are used; the temperature from the heating tank is stored in memory through MD100, and the temperature from the process tank is stored in memory through MD200. Two digital outputs control the heating element and the pump.

## 2. Lab Tasks and Design Justification

### Scaling the analogue inputs

The analogue temperature inputs are automatically converted to digital through the PLC hardware. However, the raw values have a different range and therefore need to be scaled to a range between -20°C and +125°C (operating range of the thermistor). The team initially attempted to use 'normal' and 'scale' blocks. But the scale block did not work as predicted, and the team was unsuccessful in getting the temperature within the range. Therefore, the team decided to use 'calculate' blocks to calibrate the temperature manually.

The team used an external temperature sensor to measure the temperature of the water at different levels and recorded the raw values sent to the system. Upon taking multiple readings between 28 and 32 degrees, an inversely proportional trendline was derived:

$$\text{Temperature} = -\frac{\text{Raw value}}{556} + 63.6$$

The following equation was designed into the system using two 'calculate' blocks for each input signal. The scaled temperature was then saved to memory through 'Outt1' (MD100) or 'Outt2' (MD200) directly, without the need for a 'move' block.

### Controlling the heater

The heating element in the heating tank is connected to the PLC digital output. As the maximum value in the tank is to be kept under 40°C, the logic is added using a 'less-than comparator' such that the heater turns on while the temperature (Outt1) is kept under 38°C (leeway of 2°C). Another line of logic was added to turn the heater off when the temperature goes over 40°C; however, this line may be redundant, as the heater would not turn on if the temperature is greater than 40°C to begin with.

To maximise life, the heater is to be kept on for a minimum of 5 seconds. Therefore, the logic to turn on the heater is sealed-in using the 'keepheaton variable', where a 5-second 'Timer-On delay' is initiated when the criteria to turn on the heater (temperature is below 38°C) is met, 'keepheaton' is turned on until bit 'Q' of the Timer is triggered after the 5 seconds, during which the 'keepheaton' is turned off.

### Controlling the pump

The same logic is applied to the pump. The pump is turned on when the temperature of the heating tank is greater than 30°C (to ensure the water pumped up meets the criteria of the process tank) and the temperature in the process tank is less than 28°C (to heat the water that is already in the process tank to meet the desired temperature). Like the heater, the network to

turn off the pump when the temperature is greater than 32°C may be redundant, as the pump wouldn't be turned on to begin with. The seal-in logic through the 'keepump' variable applies the same principle as that of the heater, with a 2-second minimum delay instead of 5.

### **Extra activities (not implemented)**

While the team was capable of doing the extra activities, they ran out of time to implement the logic. The task involves alerting if the pump or heating system has failed. This can be done by adding a new network where a red lamp is turned on if the following logic is true:

- *'Criteria to turn on heater/pump' AND 'heater/pump NOT turned on'.*

Additionally, the task requires the operator to set the desired temperature in the process tank to either 30°C or 27°C using the buttons on the PLC panel. This can be implemented as follows:

- *Add a new network for the 27°C setpoint where the pump turns on between the range of  $x > 25$  and  $x < 29$  (like the network already present for the 30°C setpoint).*
- *Modify both the 30°C and 27°C setpoint networks by adding another normally open contact, which is tagged to the digital input of a button on the PLC panel.*

Upon wiring the hardware, the new contact (one of the buttons) will now determine which set point is active. For this scenario, the contact is assumed to be pushed down to avoid the complexity of adding another seal-in.

## **3. Reflection & Conclusion**

### **Design limitations and challenges**

The main tasks were completed successfully and demonstrated to the tutor. However, due to the lack of experience in using the Siemens software and designing ladder logic, the team took the whole lab to complete the main tasks and was unable to implement the extra activities. Additionally, two of the networks seem to be redundant and can be removed from the program to improve simplicity and efficiency. The tag names such as 'Outt1', 'Outt2' and temporary variables can be renamed for clarification, making it easier for an outsider to interpret the logic.

Ideally, the students would have 'normalised' the analogue input and then 'scaled' it using the respective blocks. However, the team had trouble calibrating these two blocks and upon multiple failed attempts, used a 'calculate' block instead. Although the system works, the 'calculate' block is limited to the specific sensor and its conditions. It will need to be entirely recalibrated if the conditions change.

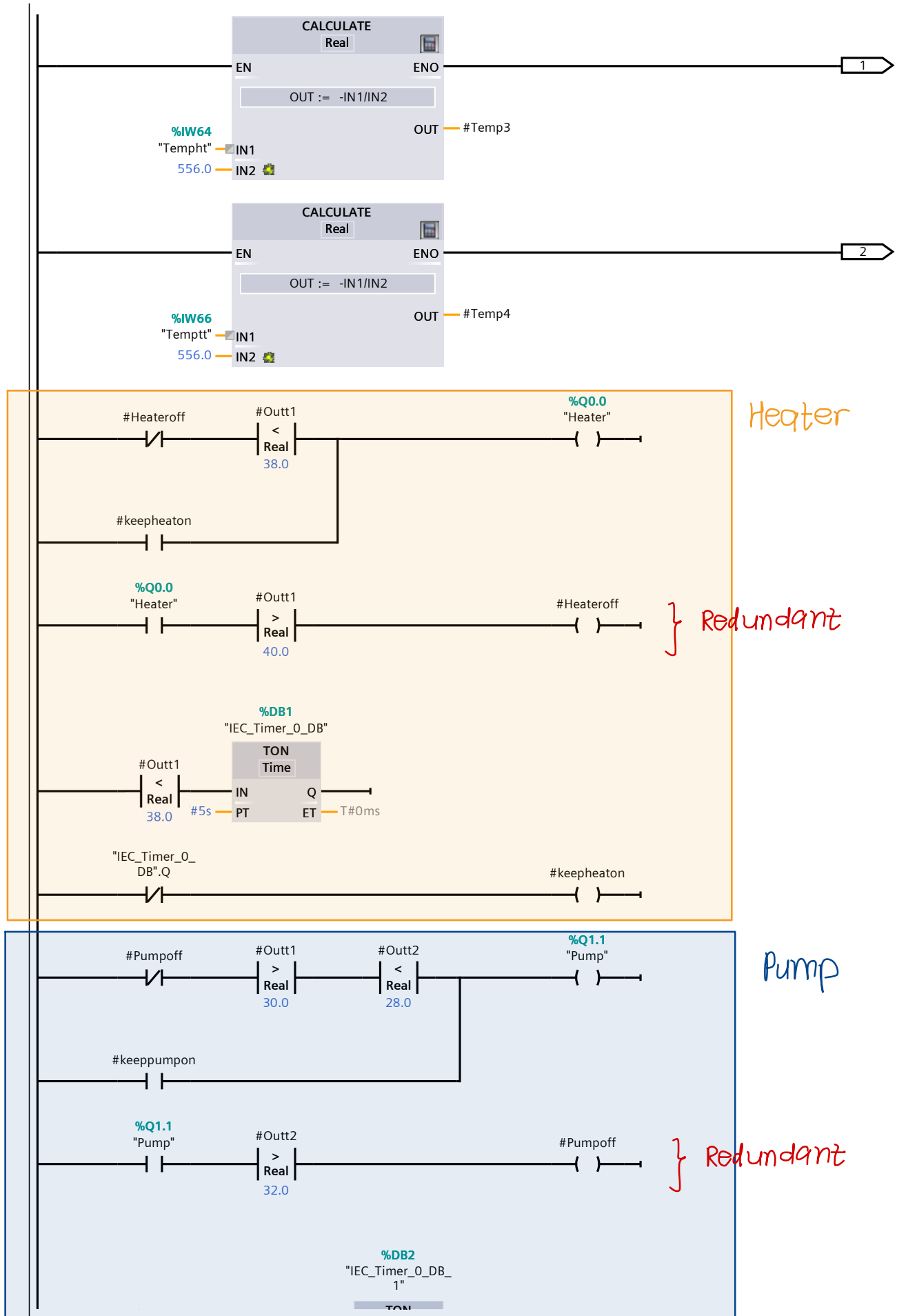
The team had challenges in implementing the seal-in features for both the heater and pump. Testing if the heater worked was very time-consuming as the thermistor operated very slowly. It was also observed that due to this inefficient thermistor, the pump was turned on and off every 2 seconds (within minimum duration) continuously at boundary temperatures such as 30°C, until the thermistor eventually passed this temperature. Difficulties were also faced during testing, as the students had to wait long periods until the water cooled back down; however, this was later avoided by adding chilled water.

### **Lessons learned**

In future labs, more preparation is required as part of the prelab to avoid the unnecessary time spent during the lab. The final section of the lab is to be dedicated to refining the code and making it more readable. Nevertheless, the group was successful in completing the main tasks and have enhanced their understanding of designing ladder logic diagrams.

## 4. Appendix

### A. Ladder Logic Diagram



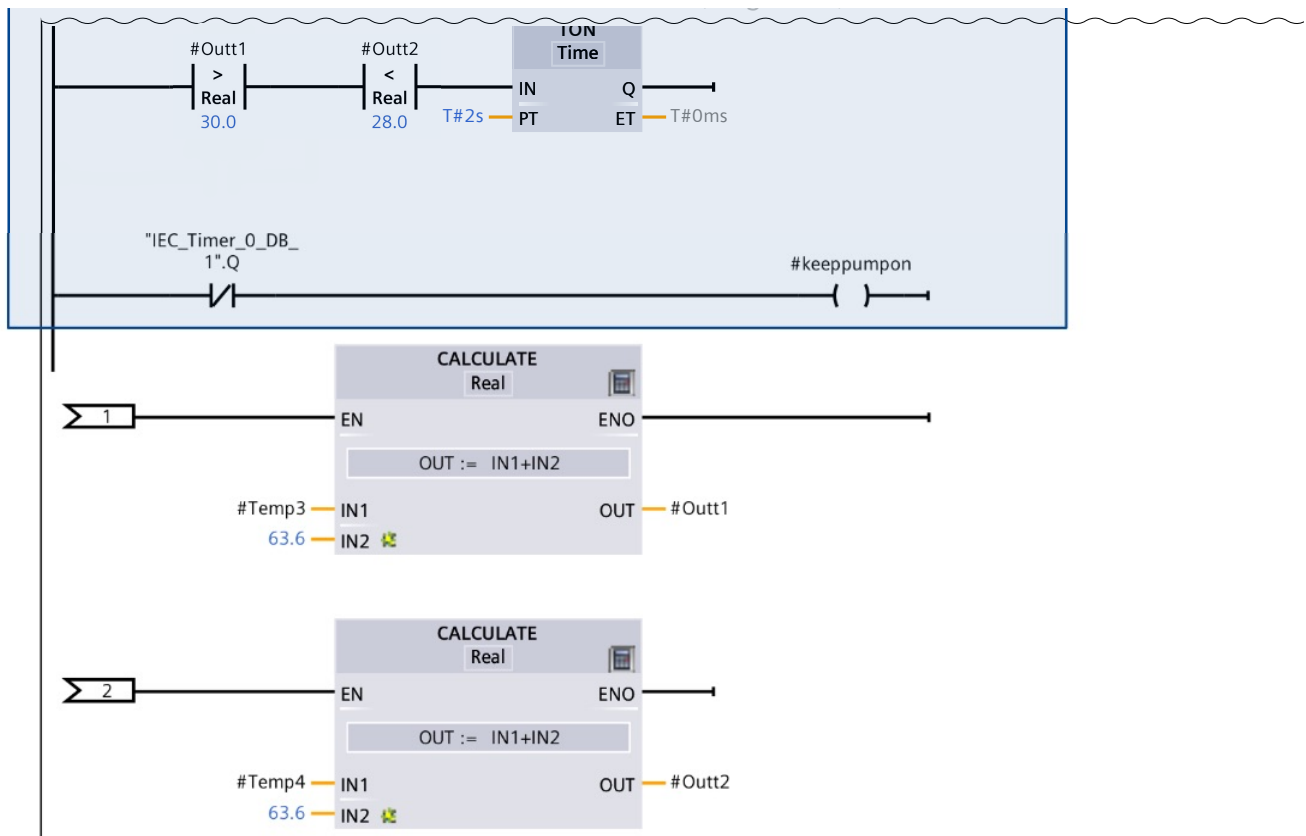


Table 1: Description for tags used in the ladder logic diagram

| Tag Name         | Description                                         |
|------------------|-----------------------------------------------------|
| Heater           | Turn on heater                                      |
| Heateroff        | Turn off heater                                     |
| IEC Timer 0 DB   | 5s delay timer                                      |
| IEC Timer 0 DB 1 | 2s delay timer                                      |
| keepheaton       | Seal-in heater on                                   |
| keepumpmon       | Seal-in pump on                                     |
| Outt1 (MD100)    | Temperature of heating tank (scaled)                |
| Outt2 (MD200)    | Temperature of process tank (scaled)                |
| Pump             | Turn on pump                                        |
| Pumpoff          | Turn off pump                                       |
| Temp3            | Temporary variable to scale heating tank (gradient) |
| Temp4            | Temporary variable to scale process tank (gradient) |
| Tempht           | Temperature of heating tank (analogue, unscaled)    |
| Temptt           | Temperature of process tank (analogue, unscaled)    |